OBSTRUCTIVE SLEEP APNEA



Obstructive sleep apnea (OSA) is a chronic medical condition that is commonly undiagnosed. The most common form of sleep-disordered breathing, it is characterized by recurring transient obstructions of airflow that occur exclusively during sleep. OSA is associated with increased risk of perioperative complications, including postoperative respiratory failure, cardiac events, and ICU transfer, and should be identified and treated as early as possible to help reduce this risk. [1-7]

Screening Tools

The STOP-Bang questionnaire is a validated tool for preoperative screening for OSA, assessing the likelihood of moderate to severe OSA. Scores of 5 or higher indicate a high probability of moderate to severe OSA. [8]

STOP-Bang Questionnaire

STOP-Bang Questionnaire



Is it possible that you have ...
Obstructive Sleep Apnea (OSA)?

Please answer the following questions below to determine if you might be at risk.

Yes ○ No ○ **Snoring ?**

Do you **Snore Loudly** (loud enough to be heard through closed doors or your bed-partner elbows you for snoring at night)?

Yes ○ No ○ **Tired ?**

Do you often feel **Tired, Fatigued, or Sleepy** during the daytime (such as falling asleep during driving or talking to someone)?

Yes ○ No ○ **Observed ?**

Has anyone **Observed** you **Stop Breathing** or **Choking/Gasping** during your sleep ?

Yes ○ No ○ **Pressure ?**

Do you have or are being treated for **High Blood Pressure** ?

Yes ○ No ○ **Body Mass Index more than 35 kg/m^2?**

Body Mass Index Calculator
○ cm / kg ○ inches / lb

Height: Weight:

Calculate

BMI:

Yes ○ No ○ **Age older than 50 ?**

Yes ○ No ○ **Neck size large ? (Measured around Adams apple)**

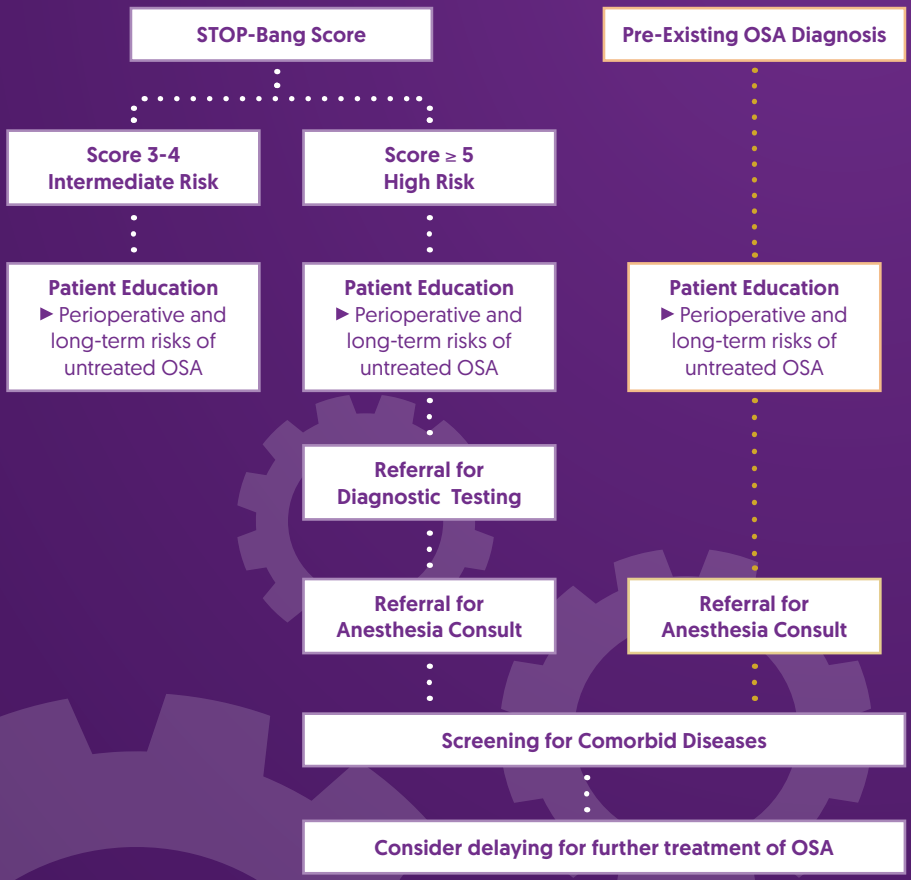
Is your shirt collar 16 inches / 40cm or larger?

Yes ○ No ○ **Gender = Male ?**

See Result

For general population
OSA - Low Risk : Yes to 0 - 2 questions
OSA - Intermediate Risk : Yes to 3 - 4 questions
OSA - High Risk : Yes to 5 - 8 questions
or Yes to 2 or more of 4 STOP questions + male gender
or Yes to 2 or more of 4 STOP questions + BMI > 35kg/m^2
or Yes to 2 or more of 4 STOP questions + neck circumference 16 inches / 40cm

OBSTRUCTIVE SLEEP APNEA ALGORITHM



Patient Education

- Counsel patients on the perioperative and long-term risks of untreated moderate or severe obstructive sleep apnea
- For pre-existing OSA diagnosis:
  - Instruct patient to continue using their treatment device (e.g., CPAP, BPAP, or dental device) until their surgery, and bring it to hospital, including for daycare surgery
  - Request sleep study results from primary care provider if not available on electronic health record
  - If patient is not using their treatment device as prescribed, emphasize the importance of its use in the postoperative period and suggest they follow up with their family doctor or device vendor to discuss barriers to use and potential solutions

Referral for Diagnostic Testing

- Home Sleep Apnea Test (HSAT) (without Sleep Disorder Physician consultation)
  - See Form A for inclusion and exclusion criteria. This testing should be ordered by a physician who is responsible for the longitudinal care of a diagnosis of sleep apnea.
- Sleep Disorder Consultation Referral
  - Sleep physicians will determine the appropriate testing modality and follow for therapy
  - Use for patients that are not appropriate for HSAT
- See online provider resources for referral templates

Prehabilitation and Optimization Recommendations

Referral for Anesthesia Consult	⚙ Consider referral in patients coming for moderate to severely invasive surgery, airway surgery, or those with severe OSA not compliant with CPAP ⚙ To plan for postoperative management & monitoring to minimize OSA related perioperative complications
Screening for Comorbid Disease	⚙ Obesity Hypoventilation Syndrome *(see Obesity)* ⚙ Diabetes ⚙ Cardiovascular / Respiratory complications
Consider delaying for further treatment of OSA	⚙ Made on a case by case basis taking into consideration: • Patient's overall health • Urgency of the surgery • Potential impact of untreated OSA on the surgical outcome ⚙ In the vast majority of cases there is insufficient evidence to support delaying surgery for testing/ treatment unless there is evidence of significant or uncontrolled systemic disease or additional problems with ventilation or gas exchange. Continuation with surgery is reasonable if the patient is managed perioperatively as though they have untreated moderate to severe OSA [9]